Practical 3: MongoDB Operations

Aim: To perform single and multiple document insertion and retrieval.

Code :-

```
STUDENT_NAME = "Prathama Dinesh Gaikwad"
# Simulated collections
items = []
products = []
# Step 1: Insert Single Document
print("\n## Step 1: Insert Single Document")
item = {"name": "laptop", "price": 999}
items.append(item)
print("Inserted document:")
print(item)
# Step 2: View Inserted Document
print("\n## Step 2: View Inserted Document")
for document in items:
    print(document)
# Step 3: Insert Multiple Documents
print("\n## Step 3: Insert Multiple Documents")
new_products = [
    {"name": "phone", "price": 500, "stock": 10},
    {"name": "tablet", "price": 300, "stock": 5},
    {"name": "watch", "price": 150, "stock": 0}
]
products.extend(new_products)
print("Inserted products:")
for product in products:
    print(product)
```

```
# Step 4: View Products in Formatted JSON
print("\n## Step 4: View Products")
import json
print(json.dumps(products, indent=4))
# Display student name
print("\n========================================")
print("Submitted by:", STUDENT_NAME)
print("========================================")
print("Hello, World!")
```

Output:-

```
## Step 1: Insert Single Document
Inserted document:
{'name': 'laptop', 'price': 999}

## Step 2: View Inserted Document
{'name': 'laptop', 'price': 999}

## Step 3: Insert Multiple Documents
Inserted products:
{'name': 'phone', 'price': 500, 'stock': 10}
{'name': 'tablet', 'price': 300, 'stock': 5}
{'name': 'watch', 'price': 150, 'stock': 0}

## Step 4: View Products
[
    {
        "name": "phone",
        "price": 500,
        "stock": 10
    },
    {
        "name": "tablet",
         "price": 300,
         "stock": 5
     },
     {
         "name": "watch",
         "price": 150,
         "stock": 0
     }
]

========================================
Submitted by: Prathama Dinesh Gaikwad
========================================
Hello, World!
```